

Probabilistic Graphical Models

CSE 5830

Lecture 3

Bayesian Network Representation

Goals for Today

- Define Bayesian Networks
- Naive Bayes
- Relation between BNs and independence
- V-structure, active trail,
- D-separation, I-Map

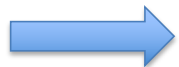
Why Bayesian Networks?

Goal:

Represent a joint distribution P over some set of random variables $\mathcal{X} = \{X_1, \dots, X_n\}$

The explicit representation is not practical

- Computationally, very expensive (memory)
 - E.g. For n binary random variables we need $2^n - 1$ parameter
- Cognitively, can not be handled
- Statistically, not enough samples to estimate parameters



Exploiting independence properties

Exploiting independence properties (revisited)

The explicit representation of a joint distribution is not practical

Bayesian Network:

- Representation of independence properties of the distribution
- Use of alternative parameterization that allows exploiting of these independencies

Independent random variable

- Without considering independency for n binary random variables we need $2^n - 1$ parameters to represent the distribution
- n independent random variables: $X_1, \dots, X_n$
 - $X_i \perp X_j \Rightarrow P(X_1, \dots, X_n) = P(X_1)P(X_2) \cdots P(X_n)$

Parameterization
 - We need n parameters to represent distribution
 - Example tossing coin, θ_i :prob(head) for coin i

Conditional parameterization, revisited

- In real word problems, random variables are not independent
- Conditional parameterization provides a basis for compact representation of complex distribution

Example: probability of intelligence and SAT score of a student
Assuming binary random variables I (intelligence) and S (student)
 $P(S,I)$

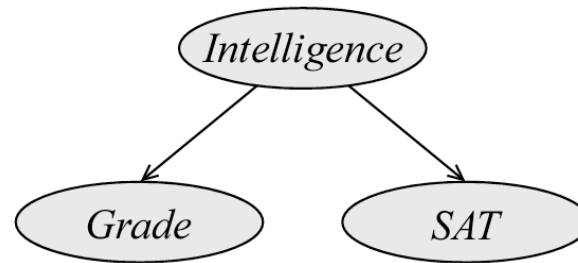


S depends on I or
I causes S

Conditional independency → causality → Bayesian Network

Example of simple BN

- Intelligence (I)
 i^0 (low), i^1 (high),
- SAT (S)
 s^0 (low), s^1 (high)
- Grade (G)
 g^1 (A), g^2 (B), g^3 (C)



$$P(I, S, G)$$

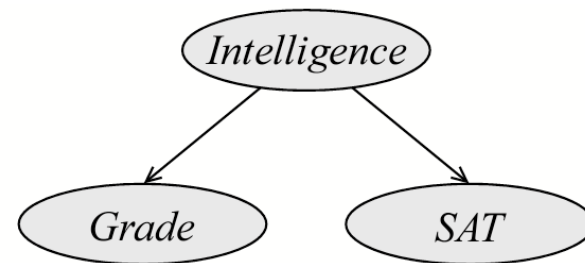
$$P \models (S \perp G \mid I)$$

Compact specification of the joint distribution

$$P \models (S \perp G \mid I)$$

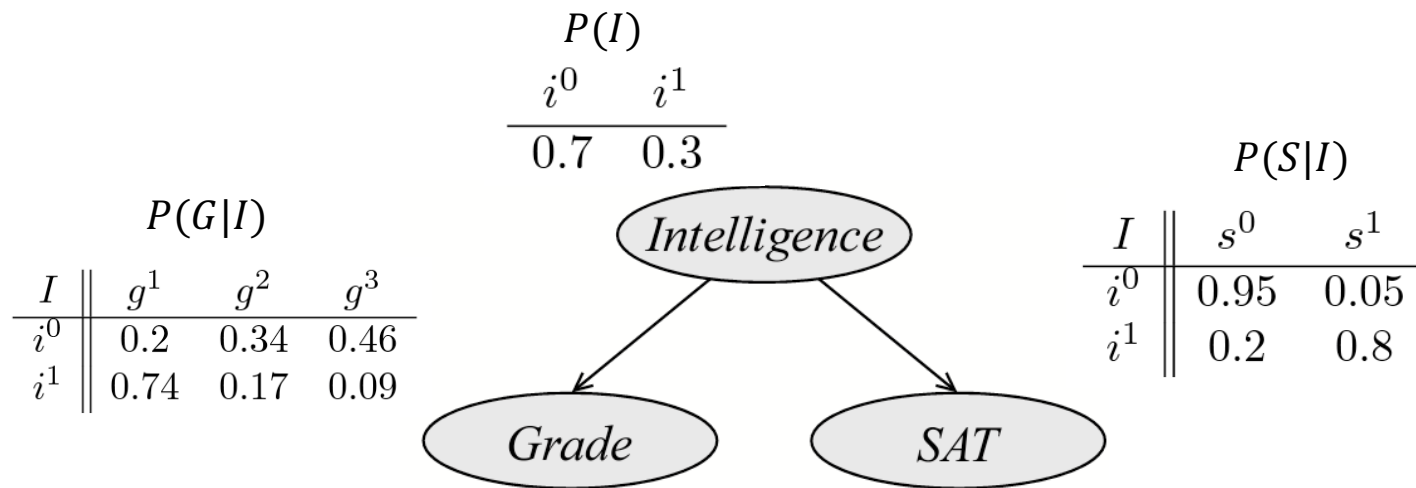
$$P(I, S, G) = P(S, G \mid I)P(I)$$

$$P(S, G \mid I) = P(S \mid I)P(G \mid I)$$



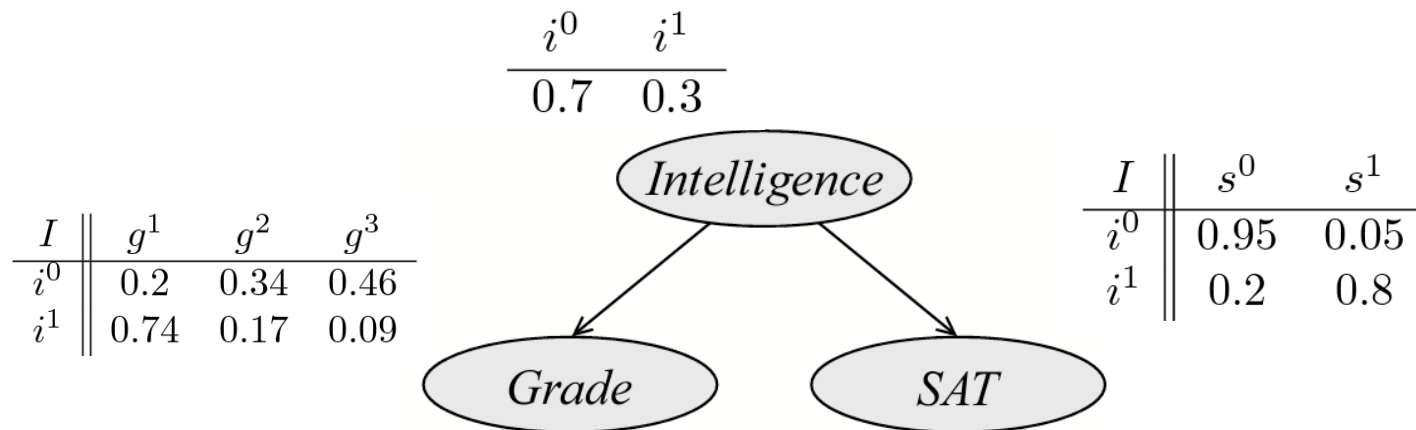
$$P(I, S, G) = P(S \mid I)P(G \mid I)P(I)$$

Example of simple BN



$$P(i^1, s^1, g^2)$$

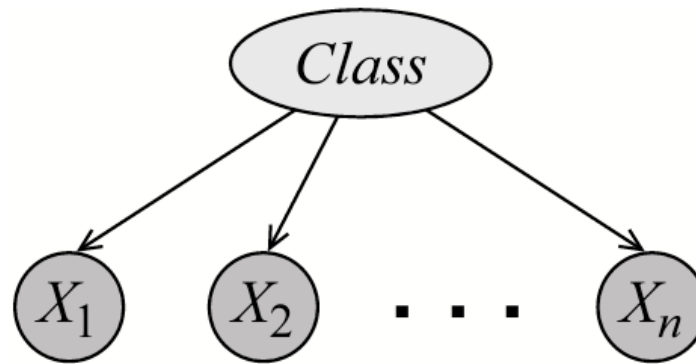
Simple Naïve Bayes model



$$\begin{aligned}
 P(i^1, s^1, g^2) &= P(i^1)P(s^1 | i^1)P(g^2 | i^1) \\
 &= 0.3 \cdot 0.8 \cdot 0.17 = 0.0408.
 \end{aligned}$$

How many parameters we need using the conditional parameterization?
 How many parameters we need using the joint probability?

The Naïve Bayes Model



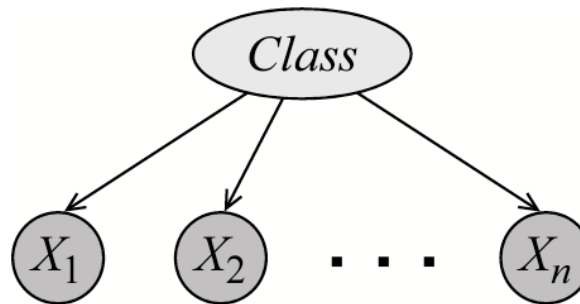
Observed features $X_1, \dots, X_n$ are conditionally independent

$$(X_i \perp \mathbf{X}_{-i} \mid C) \quad \text{for all } i, \quad \text{where } \mathbf{X}_{-i} = \{X_1, \dots, X_n\} - \{X_i\}$$

Model factorization

$$P(C, X_1, \dots, X_n) = P(C) \prod_{i=1}^n P(X_i \mid C)$$

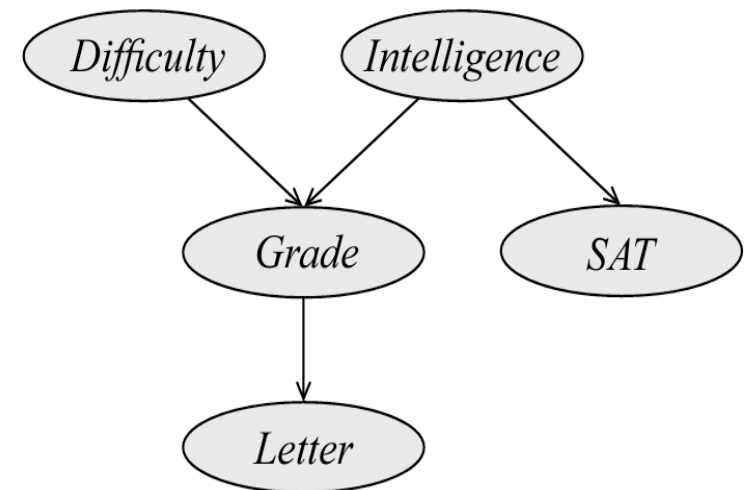
The Naïve Bayes Model for Classification



$$\frac{P(C = c^1 \mid x_1, \dots, x_n)}{P(C = c^2 \mid x_1, \dots, x_n)} = \frac{P(C = c^1)}{P(C = c^2)} \prod_{i=1}^n \frac{P(x_i \mid C = c^1)}{P(x_i \mid C = c^2)};$$

Bayesian network

- A BN is a directed graph whose nodes represent the random variables and whose edges represent direct influence of one variable on another
- It is a data structure that provides the skeleton for representing a joint distribution compactly in a factorized way
- It offers a compact representation for a set of conditional independence assumptions about a distribution



The Bayesian Network Independence Assumption

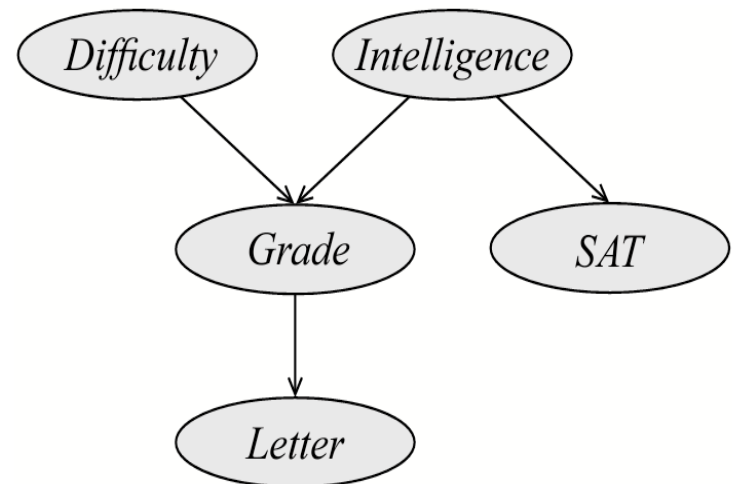
Local Markov Assumption: A variable X is independent of its non-descendants given its parents (and *only* its parents)

$$X \perp \mathbf{NonDescendants}(X) \mid \mathbf{Parents}(X)$$

- P “factorizes over graph G ” defined by $\mathbf{Parents}()$

The Bayesian Network Independence Assumption

Example



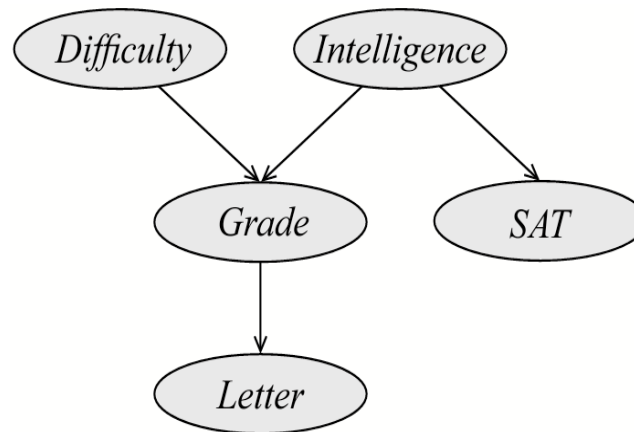
Formal definition of a BN

- A BN structure G encodes local independencies
- For each random variable X_i :
 $(X_i \perp \text{NonDescendants}(X_i) \mid \text{Parents}(X_i))$
- Joint distribution factorizes over G

$$P(\mathbf{X}) = \prod_{i=1}^n P(X_i \mid \mathbf{Parents}(X_i))$$

P factorizes over G

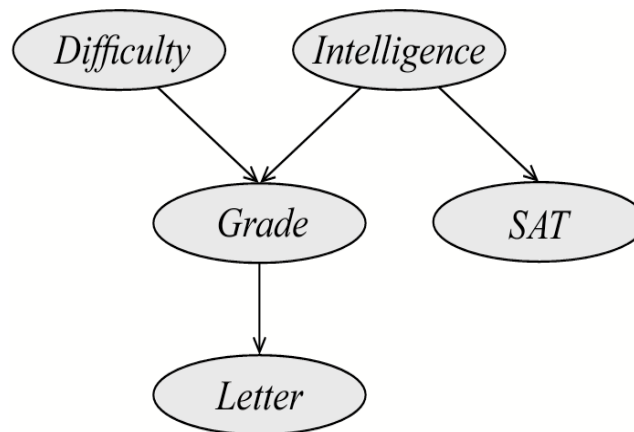
$$P(\mathbf{X}) = \prod_{i=1}^n P(X_i \mid \mathbf{Parents}(X_i))$$



$$P(I, D, G, S, L) = .$$

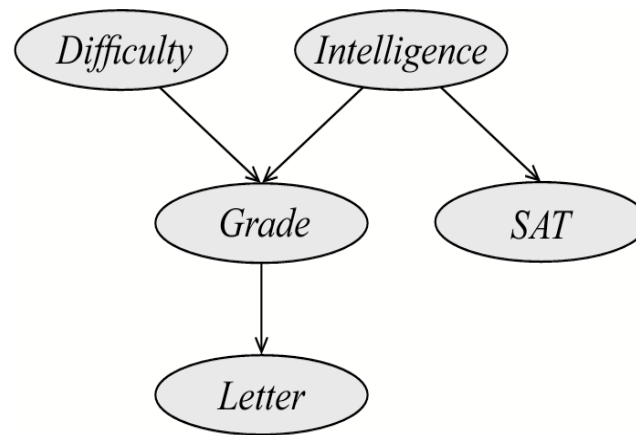
P factorizes over G

$$P(\mathbf{X}) = \prod_{i=1}^n P(X_i \mid \mathbf{Parents}(X_i))$$

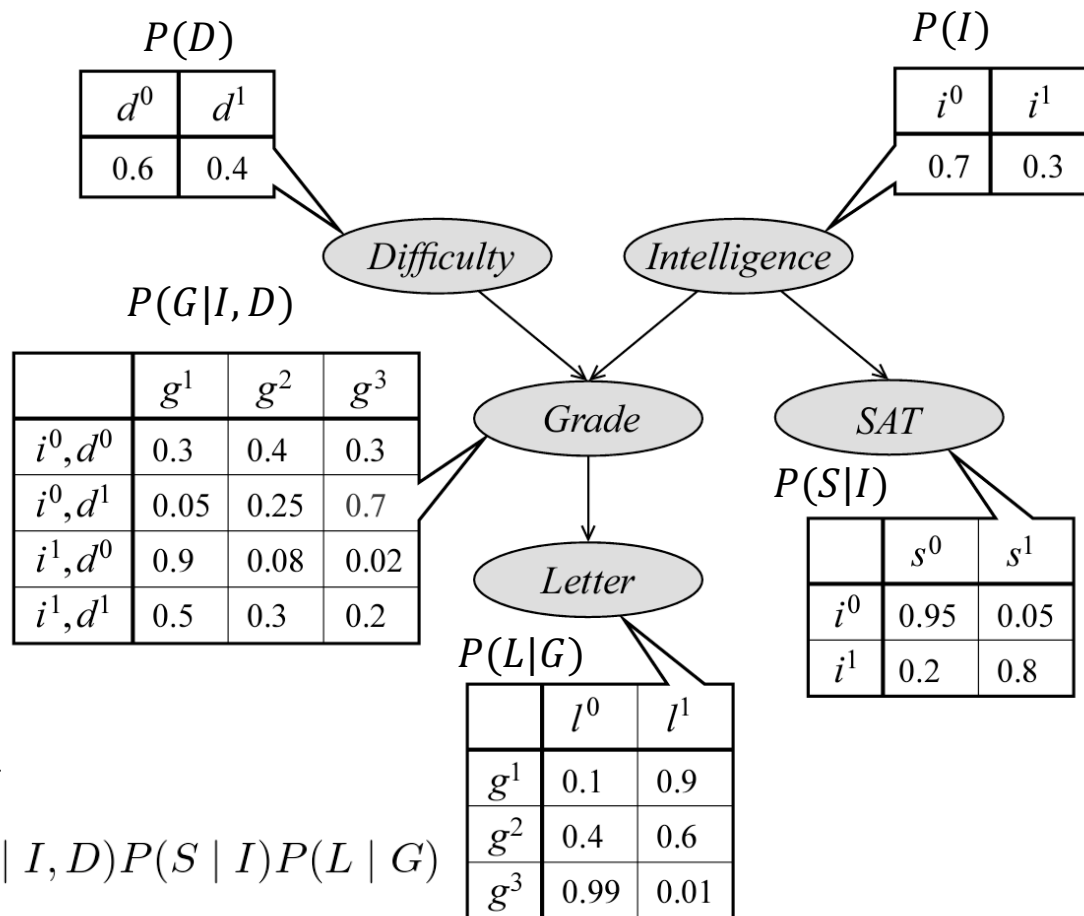


$$P(I, D, G, S, L) = P(I)P(D)P(G \mid I, D)P(S \mid I)P(L \mid G)$$

Chain rule for Bayesian networks



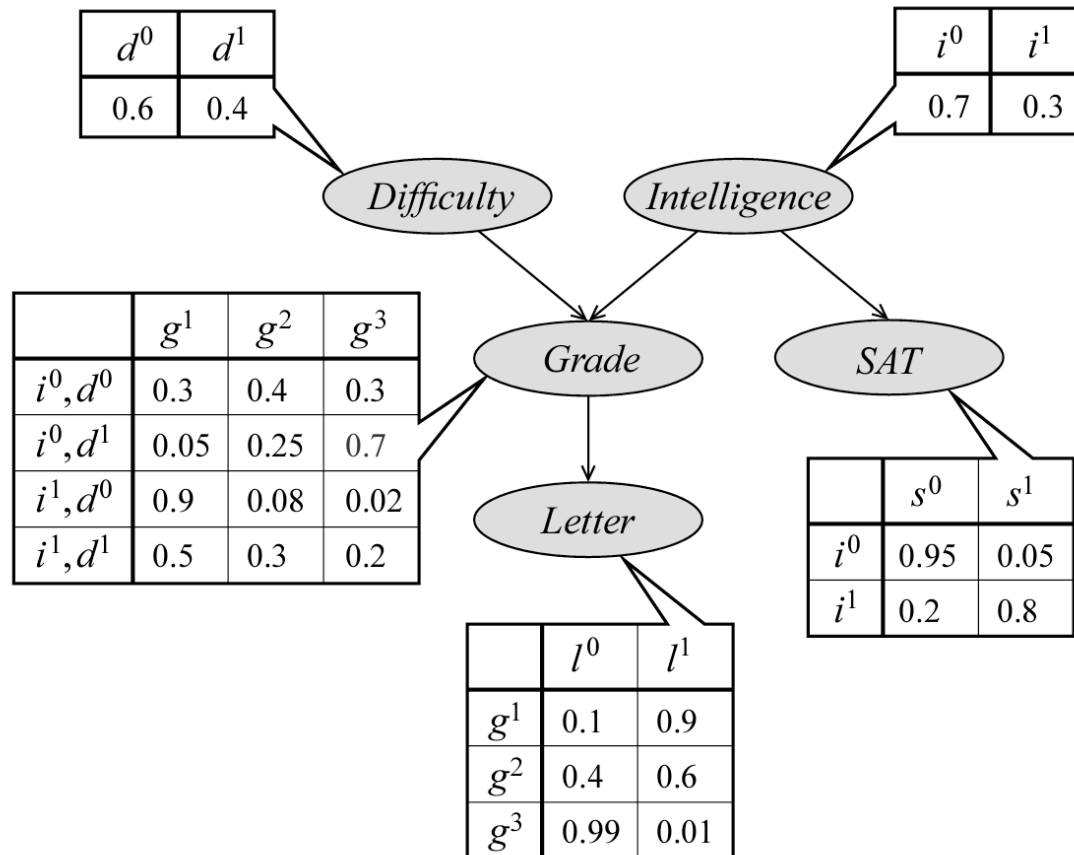
$$P(I, D, G, S, L) = P(I)P(D)P(G \mid I, D)P(S \mid I)P(L \mid G)$$



$$P(I, D, G, S, L) = .$$

$$P(I)P(D)P(G | I, D)P(S | I)P(L | G)$$

$$P(i^1, d^0, g^2, s^1, l^0)$$



$$\begin{aligned}
 P(i^1, d^0, g^2, s^1, l^0) &= P(i^1)P(d^0)P(g^2 \mid i^1, d^0)P(s^1 \mid i^1)P(l^0 \mid g^2) \\
 &= 0.3 \cdot 0.6 \cdot 0.08 \cdot 0.8 \cdot 0.4 = 0.004608.
 \end{aligned}$$

BN Is a Legal Distribution:

$$\sum P = 1, P \geq 0$$

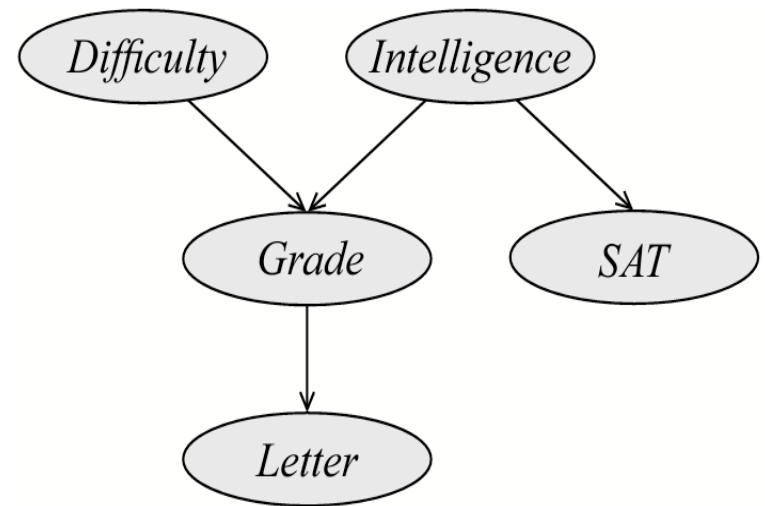
- P is product of CPDs and CPDs are non-negative $\rightarrow P \geq 0$

- $\sum P = 1$:

$$\begin{aligned}\sum_{D,I,G,S,L} P(D,I,G,S,L) &= \sum_{D,I,G,S,L} P(D) P(I) P(G|I,D) P(S|I) P(L|G) \\ &= \sum_{D,I,G,S} P(D) P(I) P(G|I,D) P(S|I) \sum_L P(L|G) \\ &= \sum_{D,I,G,S} P(D) P(I) P(G|I,D) P(S|I) \\ &= \sum_{D,I,G} P(D) P(I) P(G|I,D) \sum_S P(S|I) \\ &= \sum_{D,I} P(D) P(I) \sum_G P(G|I,D) \\ &= 1\end{aligned}$$

Queries and Reasoning Patterns

- Causal Reasoning or Prediction (downstream)
 - E.g. $P(L|I)$
- Evidential Reasoning (upstream)
 - E.g. $P(D|G)$
- Inter-causal Reasoning (sideways between parents)
“explaining away”
 - E.g. $P(I|G,D)$



Reading Independencies from the Graph

Flow of probability influence

When can X influence Y ?

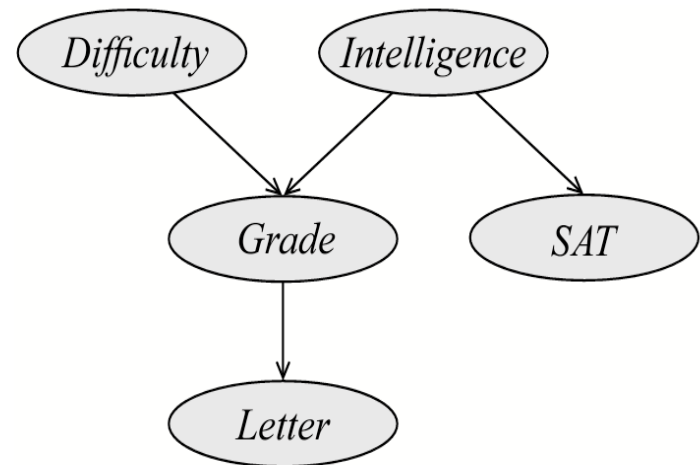
Condition on X changes our belief about Y

Are X and Y independent?

- $X \rightarrow Y$ $\times$
- $X \leftarrow Y$ $\times$
- $X \rightarrow W \rightarrow Y$ $\times$
- $X \leftarrow W \leftarrow Y$ $\times$
- $X \leftarrow W \rightarrow Y$ $\times$
- $X \rightarrow W \leftarrow Y$ $\checkmark$

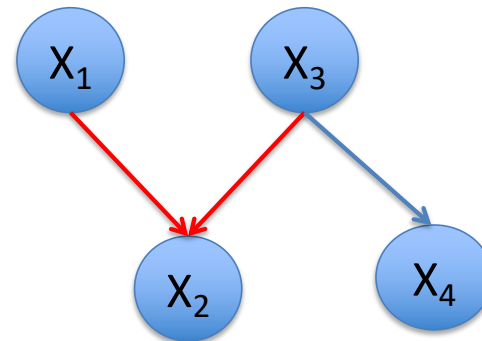
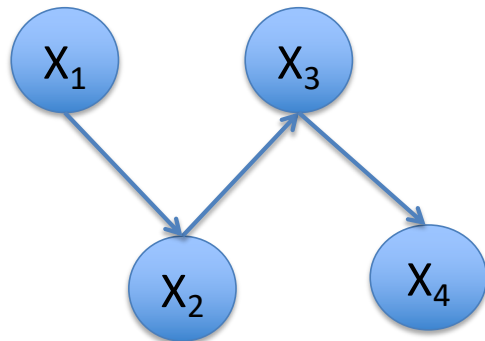


V-structure



Active Trails

- Let G be a BN structure :
- A trail $X_1 - \dots - X_k$ in G is active if: it has no v-structures
 $X_{i-1} \rightarrow X_i \leftarrow X_{i+1}$
 - Influence can flow, unless it is block by a v-structure



“Active trail” $\sim$ dependent

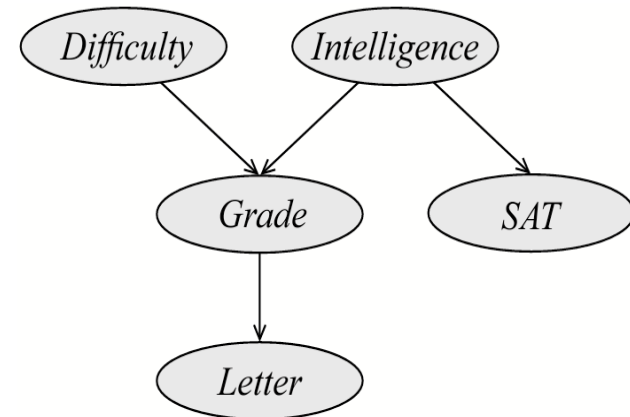
Reading Independencies from the Graph with evidence

When can X influence Y Given evidence about Z
 Are X and Y independent?

- $X \rightarrow Y$ X
- $X \leftarrow Y$ X
- $X \rightarrow W \rightarrow Y$
- $X \leftarrow W \leftarrow Y$
- $X \leftarrow W \rightarrow Y$
- $X \rightarrow W \leftarrow Y$

V-structure

	$W \notin Z$	$W \in Z$
$X \rightarrow W \rightarrow Y$	X	$\checkmark$
$X \leftarrow W \leftarrow Y$	X	$\checkmark$
$X \leftarrow W \rightarrow Y$	X	$\checkmark$
$X \rightarrow W \leftarrow Y$	$\checkmark$	X



Active Trails

Let G be a BN structure :

A trail $X_1 - \dots - X_k$ in G given Z is active if:

- for any v-structure $X_{i-1} \rightarrow X_i \leftarrow X_{i+1}$ we have that X_i or one of its descendants $\in Z$
- no other X_i is in Z

(no blocking the flow of information, passing information)

“Active trail” $\sim$ dependent

Reading Independencies from the Graph

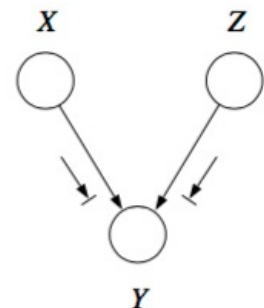
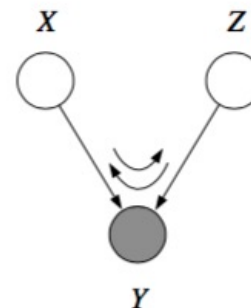
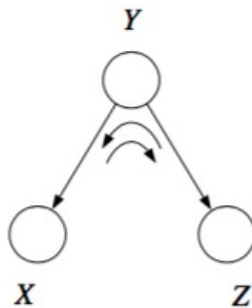
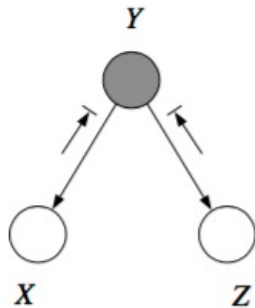
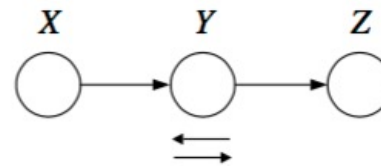
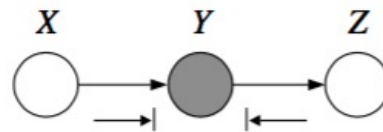
	<u>$\neg X \perp Y$</u>	<u>$X \perp Y$</u>
Direct Connection $X \rightarrow Y$		
Indirect Causal Effect $X \rightarrow Z \rightarrow Y$		
Indirect Evidential Effect $X \leftarrow Z \leftarrow Y$		
Common Cause $X \leftarrow Z \rightarrow Y$		
Common Effect $X \rightarrow Z \leftarrow Y$		

"V-structure"
"explaining away" or an observed descendant!

Bayes Ball Algorithm

Another expression of “active trails” and d-separation.

Behavior going from X to Z on path through Y.



D-Separation

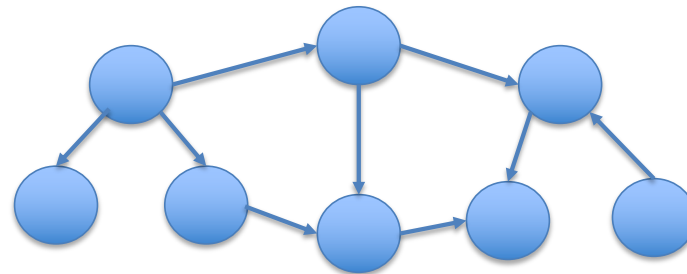
“Directed Separation”

Let X, Y, Z be three sets of nodes in G .

We say that X and Y are “*d-separated*” given Z ,

$$\text{d-sep}_G(X; Y \mid Z),$$

if there is no “*active trail*” between any node $X \in X$ and $Y \in Y$ given Z .



“Separated” $\sim$ independent

D-Separation and Dependencies

- Theorem 3.4, (K&F p73):
Let G be a BN structure. If X and Y are not d-separated given Z in G , then X and Y are dependent given Z in some distribution P that factorizes over G .
- We use $I(G)$ to denote the set of independencies that correspond to d-separation:
$$I(G) = \{ (X \perp Y \mid Z) : d\text{-sep}_G(X; Y \mid Z) \}.$$
- $I(G)$ = the set of independencies guaranteed in all P_G

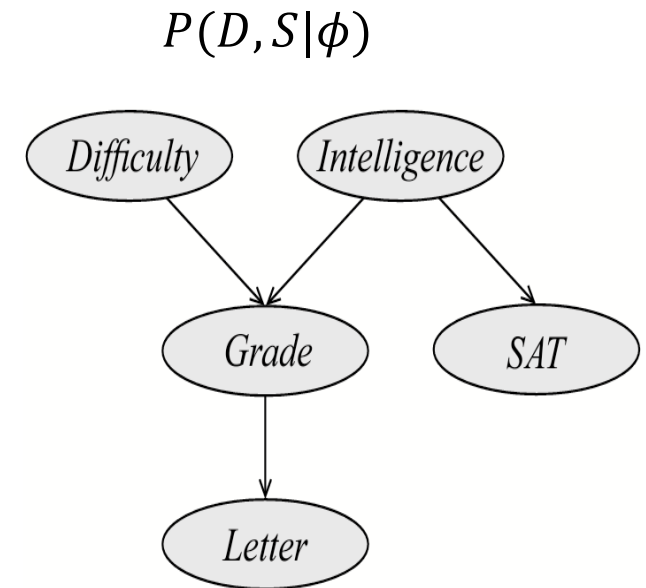
Factorization $\Rightarrow$ Independence: BNs

- If P factorizes over G , and $\text{d-sep}_G(\mathbf{X}, \mathbf{Y} \mid \mathbf{Z})$ then P satisfies $(\mathbf{X} \perp \mathbf{Y} \mid \mathbf{Z})$

Example: There is no active trail between D and S

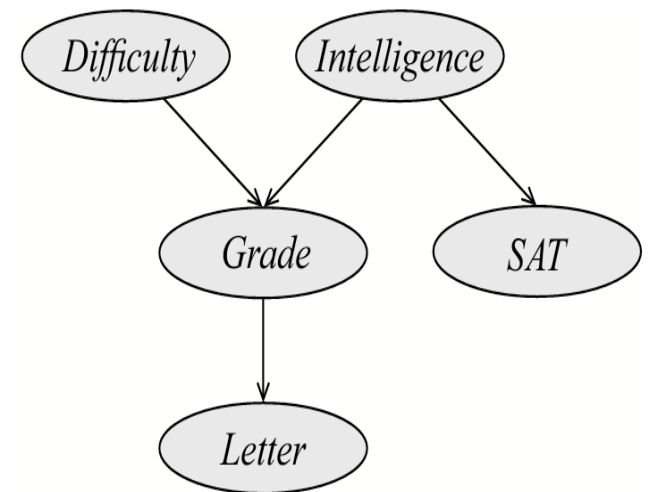
$$P(D, S \mid \phi) = P(D, S) = \sum_{I, G, L} P(D, I, G, S, L)$$

$$P(I, D, G, S, L) = P(I)P(D)P(G \mid I, D)P(S \mid I)P(L \mid G)$$



Factorization $\Rightarrow$ Independence

- Any node is d-separated from its non-descendants given its parents
- If P factorizes over G , then in P , any variable is independent of its non-descendants given its parents



I-maps

“Independency Map”

- d-separation in $G \Rightarrow P$ satisfies corresponding independence statement

$$I(G) = \{(X \perp Y | Z) : d\text{-sep}_G(X, Y | Z)\}$$

- Definition: If P satisfies $I(G)$, we say that G is an I-map (independency map) of P
- If G is an I-map for $P \Rightarrow P$ is factorized according to G

I-maps

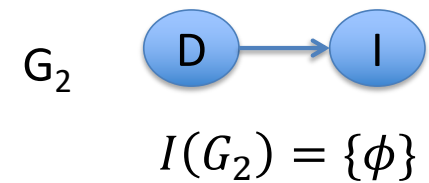
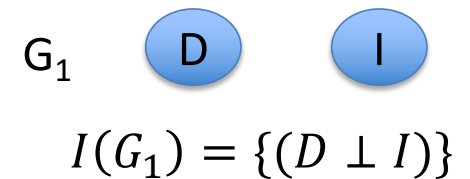
“Independency Map”

P_1

I	D	Prob
i^0	d^0	0.42
i^0	d^1	0.18
i^1	d^0	0.28
i^1	d^1	0.12

P_2

I	D	Prob.
i^0	d^0	0.282
i^0	d^1	0.02
i^1	d^0	0.564
i^1	d^1	0.134



G_1 : I-map P_1 , not an I-map for P_2

G_2 : I-map for both P_1 and P_2

Factorization & Dependency

- Factorization $\Leftrightarrow$ Independence: BNs
 - **Theorem:** If P factorizes over G , then G is an I-map for P
- Independence $\Leftrightarrow$ Factorization
 - **Theorem:** If G is an I-map for P , then P factorizes over G

Factorization & Dependency

- Two equivalent views of graph structure:
 - Factorization: G allows P to be represented
 - I-map: Independencies encoded by G hold in P
- If P factorizes over a graph G , we can read from the graph independencies that must hold in P (an independency map)